SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES SAVEETHA UNIVERSTY

Department of Computer Science and Engineering

ASSGINMENT QUESTIONS CO - 3

CSA040 OPERATING SYSTEMS

1. Consider the following string of page references

   Show the frame allocation for three frames for:

   a.FIFO

   b.LRU

   c.Optimal

   List the total number of page faults and miss rate for each policy.

   $$7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2.$$

   Answer: Page Replacement Algorithms – FIFO, LRU and Optimal

   Aim: To find the number of page faults and miss rate using FIFO, LRU and Optimal page replacement algorithms with three frames.

   Given:

   - Number of frames = 3
   - Page replacement algorithms:
     1. FIFO (First In First Out)
     2. LRU (Least Recently Used)
     3. Optimal

   Note: The page reference string is missing from your question.

   Please send the page reference string (for example: 7, 0, 1, 2, 0, 3, 0, 4).

   I'll give you:

   - Frame allocation table for FIFO
   - Frame allocation table for LRU
   - Frame allocation table for Optimal
   - Total page faults for each algorithm
   - Miss rate for each algorithm

   All in step-by-step practical record format.

   $$7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2.$$

2. Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.

   How many page faults would occur for the following replacement algorithms, assuming six, and seven frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each.

• LRU replacement

• FIFO replacement

• Optimal replacement
Answer: Page Replacement Algorithms

• Given:

• Page Reference String: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6

• Total page references = 20.

• Number of frames = 6 and 7.

• **1. Number of Page Faults**

• **Algorithm**	• **6 Frames**	• **7 Frames**
• LRU	• 7	• 7
• FIFO	• 10	• 7
• Optimal	• 7	• 7

• **2. Result**

• LRU = 7 page faults.

• FIFO = 10 and 7 page faults.

• Optimal = 7 page faults.

• Conclusion: Optimal and LRU algorithms produce the minimum number of page faults.

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3. Consider the following page reference string: 7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0,1.

Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms?

• LRU replacement

• FIFO replacement

• Optimal replacement
Answer: Page Replacement Algorithms

• Given:

• Page Reference String:

• 7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1

• Number of frames = 3

• Initially all frames are empty.

• **1. Number of Page Faults**

• **Algorithm**	• **Page Faults**
• LRU	• 18
• FIFO	• 17
• Optimal	• 13

• **2. Result**

- LRU = 18 page faults.
- FIFO = 17 page faults.
- Optimal = 13 page faults.
- Conclusion: The Optimal page replacement algorithm produces the minimum number of page faults (13).
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4. Apply the (1) FIFO, (2) LRU, and (3) optimal (OPT) replacement algorithms for the following page-reference strings:

   2, 6, 9, 2, 4, 2, 1, 7, 3, 0, 5, 2, 1, 2, 9, 5, 7, 3, 8, 5

   Indicate the number of page faults for each algorithm assuming demand paging with three frames.

   Answer: Page Replacement Algorithms

   Given:

   - Page Reference String: 2, 6, 9, 2, 4, 2, 1, 7, 3, 0, 5, 2, 1, 2, 9, 5, 7, 3, 8, 5
   - Number of frames = 3.

   Number of Page Faults

Algorithm	**Page Faults**
FIFO	18
LRU	17
Optimal	13

   Result

   Thus, the page faults are:

   - FIFO = 18
   - LRU = 17
   - Optimal = 13

   Conclusion: Optimal algorithm gives the minimum page faults (13).

5. Apply the (1) FIFO, (2) LRU, and (3) optimal replacement algorithms for the following page-reference strings: Find the number of page faults assuming four frames.

   0, 6, 3, 0, 2, 6, 3, 5, 2, 4, 1, 3, 0, 6, 1, 4, 2, 3, 5, 7

   Answer: Page Replacement Algorithms

   Given:

   - Page Reference String: 0, 6, 3, 0, 2, 6, 3, 5, 2, 4, 1, 3, 0, 6, 1, 4, 2, 3, 5, 7
   - Number of frames = 4.

   Number of Page Faults

Algorithm	**Page Faults**
FIFO	15
LRU	15
Optimal	11

   Result

   Thus, the page faults are:

   - FIFO = 15
   - LRU = 15
   - Optimal = 11

   Conclusion: Optimal algorithm gives the minimum page faults (11).